

Motion Tracking in Sports Analytics Using Computer Vision

1. Problem Understanding and Formulation

What is the Problem?

Sports analytics increasingly uses computer vision to automatically analyze the movement of athletes and sports objects during matches. Manual analysis of sports videos requires considerable time and may not provide accurate quantitative information about player movement. Therefore, an automated computer vision system is required to track athletes and sports objects from live or recorded video.

The main problem addressed in this assignment is the comparison of two computer vision approaches for motion tracking: **Optical Flow** and **Motion Estimation**. Optical Flow estimates the apparent movement of pixels or feature points between consecutive video frames and can provide detailed information about the direction and magnitude of motion. Motion Estimation determines changes or movement between frames using simpler techniques such as frame differencing or motion vectors and generally provides faster processing.

The system must achieve a suitable balance between **tracking accuracy and real-time processing performance**. Accurate tracking is important for determining athlete trajectories, movement direction, and sports-object motion, while high processing speed is necessary for live match analysis.

Expected Outcomes

The expected outcomes of the proposed system are:

- Automatic detection of motion from sports video.
- Tracking of athlete or sports-object movement.
- Estimation of displacement and motion direction.
- Generation of movement trajectories.
- Measurement of processing speed.
- Comparison of Optical Flow and Motion Estimation.
- Evaluation of accuracy, computational cost, and real-time performance.
- Selection of the most appropriate approach for sports analytics.

Information/Data Available

The primary input is a sports video consisting of a sequence of digital frames. Each frame contains pixel intensity and spatial information about athletes, sports objects, and the surrounding environment.

For a video containing N frames:

$$V = \{F_1, F_2, F_3, \dots, F_N\}$$

Each frame can be represented as:

$$F(x, y)$$

where x and y represent the pixel coordinates.

The available information includes:

- Video frames
- Frame resolution
- Frame rate
- Pixel intensity
- Feature points
- Object positions
- Motion vectors
- Frame-to-frame displacement
- Processing time

Assumptions

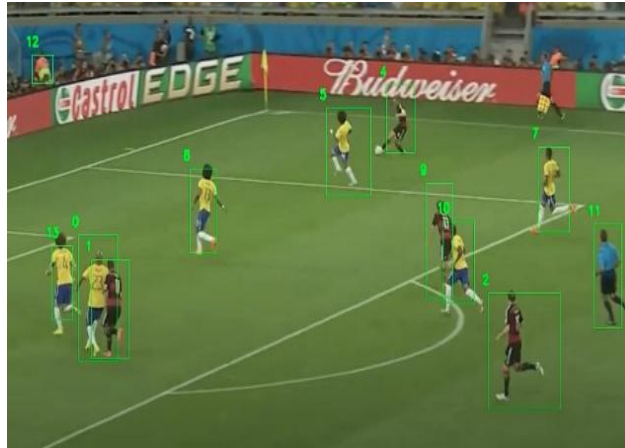
The following assumptions are made:

1. The athletes or sports objects remain visible across consecutive frames.
2. The input video has sufficient resolution for motion analysis.
3. Consecutive frames have reasonable similarity.
4. Lighting changes are not extreme.
5. The camera is fixed or has limited movement.
6. Motion between consecutive frames is measurable.
7. The video contains sufficient visual features for tracking.

Constraints

The proposed system must satisfy the following constraints:

- High tracking accuracy.
- Low processing time.
- Real-time or near-real-time operation.
- Reasonable computational requirements.
- Robustness to moderate noise and illumination changes.
- Ability to process continuous video.
- Reliable trajectory generation.
- Practical suitability for live sports analysis.



2. Application of Course Knowledge

The proposed system applies fundamental Computer Vision concepts including image representation, preprocessing, feature extraction, motion analysis, Optical Flow, Motion Estimation, and performance evaluation.

Image Representation

A video is represented as a sequence of digital images. A grayscale image can be represented by:

$$I(x, y)$$

where $I(x, y)$ represents the intensity value at pixel location (x, y) .

A color image contains three channels:

$$I(x, y) = [R(x, y), G(x, y), B(x, y)]$$

For computationally efficient motion analysis, the color frame can be converted into grayscale.

Frame Rate

The frame rate represents the number of frames processed per second.

If a video has a frame rate of FPS , the time interval between consecutive frames is:

$$\Delta t = \frac{1}{FPS}$$

For example, for a 30 FPS video:

$$\Delta t = \frac{1}{30} = 0.0333 \text{ seconds}$$

Therefore, approximately 33.3 milliseconds are available between consecutive frames for real-time processing.

Motion Estimation

Motion can be estimated by comparing two consecutive frames.

If an object moves from:

$$(x_t, y_t)$$

to:

$$(x_{t+1}, y_{t+1})$$

the displacement is:

$$\begin{aligned}\Delta x &= x_{t+1} - x_t \\ \Delta y &= y_{t+1} - y_t\end{aligned}$$

The magnitude of displacement is:

$$D = \sqrt{(\Delta x)^2 + (\Delta y)^2}$$

The approximate velocity in pixels per second is:

$$v = \frac{D}{\Delta t}$$

Optical Flow

Optical Flow estimates the apparent motion of pixels or feature points between consecutive frames.

The basic Optical Flow constraint equation is:

$$I_x u + I_y v + I_t = 0$$

where:

- I_x is the spatial intensity gradient in the x direction.
- I_y is the spatial intensity gradient in the y direction.
- I_t is the temporal intensity gradient.
- u is horizontal motion.
- v is vertical motion.

The motion vector is:

$$\mathbf{V} = (u, v)$$

and its magnitude is:

$$| \mathbf{V} | = \sqrt{u^2 + v^2}$$

Lucas-Kanade Algorithm

Lucas-Kanade Optical Flow is a feature-based Optical Flow technique. Instead of calculating motion for every pixel, it tracks selected feature points between consecutive frames.

Good feature points can be detected using the **Shi-Tomasi corner detection** method. Lucas-Kanade then estimates the new positions of those feature points.

This approach is useful for sports analytics because it provides information about the direction and magnitude of athlete movement while requiring less computation than dense Optical Flow.

Frame Differencing

Motion Estimation can be implemented using frame differencing.

For two consecutive frames:

$$D_t(x, y) = | I_t(x, y) - I_{t-1}(x, y) |$$

A threshold can then be applied to identify significant motion:

$$M(x, y) = \begin{cases} 1 & D_t(x, y) > T \\ 0 & D_t(x, y) \leq T \end{cases}$$

where T is the selected threshold.

Performance Measurement

Processing performance can be evaluated using:

$$FPS = \frac{N}{T_{total}}$$

where:

- N = number of processed frames.
- T_{total} = total processing time.

Average processing time per frame is:

$$T_{avg} = \frac{T_{total}}{N}$$

Lower processing time and higher FPS indicate better real-time performance.

If ground-truth positions are available, tracking error can be calculated as:

$$E = \sqrt{(x_p - x_g)^2 + (y_p - y_g)^2}$$

where x_p, y_p are predicted coordinates and x_g, y_g are ground-truth coordinates.

The Mean Tracking Error is:

$$MTE = \frac{1}{N} \sum_{i=1}^N E_i$$

A lower MTE indicates better tracking accuracy.



3. Solution / Design / Methodology

The proposed system follows an end-to-end computer vision pipeline:

Video Acquisition → Frame Preprocessing → Motion Analysis → Tracking
→ Trajectory Analysis → Performance Evaluation

Two approaches are implemented and compared.

Approach 1: Optical Flow

The first approach uses Lucas-Kanade Optical Flow.

Step 1: Video Acquisition

The sports video is loaded using OpenCV.

Step 2: Frame Extraction

Consecutive frames are extracted from the video.

Step 3: Preprocessing

Each frame is converted into grayscale to reduce computational complexity. Gaussian filtering can be used to reduce noise.

Step 4: Feature Detection

Shi-Tomasi corner detection is used to identify strong feature points.

Step 5: Optical Flow Calculation

Lucas-Kanade Optical Flow calculates the movement of feature points between consecutive frames.

Step 6: Trajectory Generation

The previous and current locations of feature points are connected to generate movement trajectories.

Step 7: Motion Analysis

Displacement and motion direction are calculated from the tracked feature points.

Approach 2: Motion Estimation

The second approach uses frame differencing.

Step 1: Frame Acquisition

Consecutive video frames are obtained.

Step 2: Grayscale Conversion

Frames are converted to grayscale.

Step 3: Frame Difference

The absolute difference between consecutive frames is calculated.

Step 4: Thresholding

A threshold is applied to identify significant changes.

Step 5: Morphological Processing

Morphological operations can be used to remove small noise and improve motion regions.

Step 6: Motion Region Detection

Contours are identified from the motion mask.

Step 7: Motion Visualization

Bounding boxes can be drawn around significant moving regions.

Comparison of Approaches

Parameter	Optical Flow	Motion Estimation
Motion information	Detailed	Relatively coarse
Accuracy	High	Moderate
Computational cost	Higher	Lower
Processing speed	Moderate	High
Direction information	Strong	Limited
Trajectory quality	High	Moderate
Real-time suitability	Good after optimization	Very good
Sensitivity to camera movement	Moderate	High
Suitability for detailed sports analysis	High	Moderate

Proposed Selection

The proposed solution selects **optimized Lucas-Kanade Optical Flow** because the system requires accurate motion information in addition to real-time performance.

Motion Estimation is useful when rapid detection of moving regions is the primary objective. However, sports analytics often requires more detailed trajectory and directional information. Therefore, Optical Flow provides greater analytical value.

Pseudocode

Optical Flow

START

Load sports video

Read first frame

Convert first frame to grayscale

Detect feature points using Shi-Tomasi

WHILE frames are available

 Read next frame

 Convert frame to grayscale

 Calculate Lucas-Kanade Optical Flow

 Identify new feature positions

 Remove invalid points

Calculate displacement

Draw feature points

Draw trajectories

Calculate processing time

Display tracking result

Update previous frame and points

END WHILE

Calculate average FPS

Calculate average processing time

Display results

STOP

Motion Estimation

START

Load sports video

Read first frame

Convert first frame to grayscale

WHILE frames are available

Read next frame

Convert frame to grayscale

Calculate frame difference

Apply threshold

Apply morphological filtering

Detect motion regions

Draw bounding boxes

Calculate processing time

Display result

Update previous frame

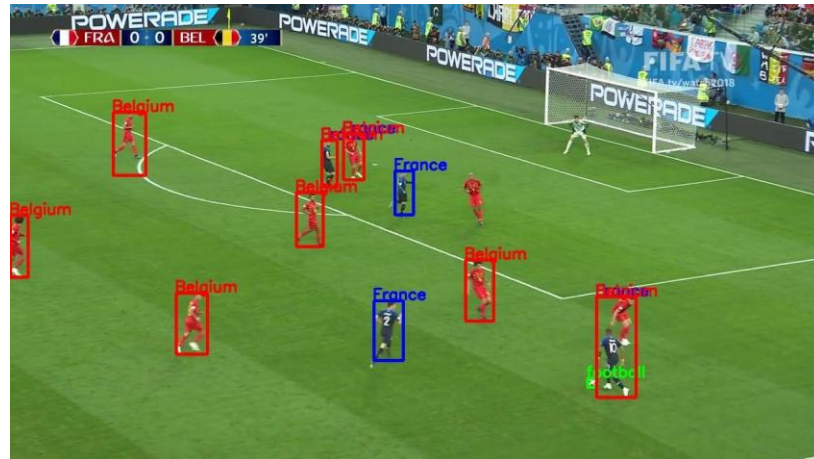
END WHILE

Calculate average FPS

Calculate average processing time

Display results

STOP



4. Use of Modern Tools

Python and OpenCV are used as the primary tools for implementing the proposed computer vision system.

Python

Python is selected because it provides a large collection of libraries for computer vision, numerical computation, and visualization. It also supports rapid development and experimentation.

OpenCV

OpenCV is used for:

- Reading sports videos.
- Extracting video frames.
- Image preprocessing.
- Grayscale conversion.
- Gaussian filtering.
- Feature detection.
- Optical Flow calculation.
- Motion estimation.
- Drawing tracking results.

- Measuring processing time.

NumPy

NumPy is used for numerical operations involving image arrays, feature coordinates, displacement calculations, and mathematical operations.

Matplotlib

Matplotlib can be used to visualize performance results such as FPS, processing time, and tracking error.

Development Environment

The implementation can be developed using:

- Visual Studio Code
- Jupyter Notebook
- Google Colab

Git and GitHub

Git is used for version control and GitHub is used to store the source code, documentation, pseudocode, video information, and experimental results.

Tool Evidence

The following evidence can be captured during implementation:

1. Python/OpenCV source code.
2. Sports video input.
3. Optical Flow tracking output.
4. Motion Estimation output.
5. Terminal showing processing results.
6. Performance graphs.
7. GitHub repository containing the complete project.

Sports Technique Analysis Using AI Pose Estimation



5. Results and Validation

The proposed system is validated by executing both Optical Flow and Motion Estimation on the same sports video.

The following parameters should be recorded during experimentation:

Metric	Optical Flow	Motion Estimation
Number of frames processed	Actual result	Actual result
Average FPS	Actual result	Actual result
Average processing time/frame	Actual result	Actual result
Number of tracked features	Actual result	Not applicable
Tracking error	Actual result	Actual result if ground truth is available
Trajectory quality	High/Medium/Low	High/Medium/Low

The numerical values must be obtained from the actual execution of the programs.

FPS Evaluation

FPS is calculated as:

$$FPS = \frac{\text{Number of Frames Processed}}{\text{Total Processing Time}}$$

A higher FPS indicates better real-time performance.

Processing Time Evaluation

Average processing time is calculated using:

$$T_{avg} = \frac{T_{total}}{N}$$

A lower value indicates faster processing.

Tracking Accuracy

When ground-truth coordinates are available, Mean Tracking Error can be calculated:

$$MTE = \frac{1}{N} \sum_{i=1}^N \sqrt{(x_{pi} - x_{gi})^2 + (y_{pi} - y_{gi})^2}$$

A lower MTE indicates better tracking accuracy.

Visual Validation

Optical Flow results should display tracked feature points and trajectories over the athlete or sports object.

Motion Estimation results should display detected moving regions or bounding boxes.

Expected Observation

The expected qualitative observation is that Optical Flow provides more detailed motion information and smoother trajectories, whereas Motion Estimation provides faster detection of moving regions with lower computational requirements.

The final experimental decision should be based on the actual FPS, processing time, tracking quality, and tracking error obtained from both implementations.

6. Analysis and Engineering Decision

The two approaches provide different advantages.

Optical Flow provides detailed information about the direction and magnitude of motion. This makes it useful for applications requiring accurate athlete trajectories, sports-object movement, and detailed motion analysis. However, Optical Flow requires more computational processing than simple frame differencing.

Motion Estimation using frame differencing is computationally simpler and generally provides faster processing. It is suitable when the main requirement is to quickly identify whether and where motion has occurred. However, it provides less detailed information about the exact movement of individual features.

The main engineering trade-off can therefore be represented as:

$$\boxed{\text{Tracking Accuracy} \leftrightarrow \text{Processing Speed}}$$

If maximum processing speed is the primary requirement, Motion Estimation is advantageous. If detailed motion information is more important, Optical Flow is preferable.

For the proposed sports analytics application, **optimized Lucas-Kanade Optical Flow is selected.**

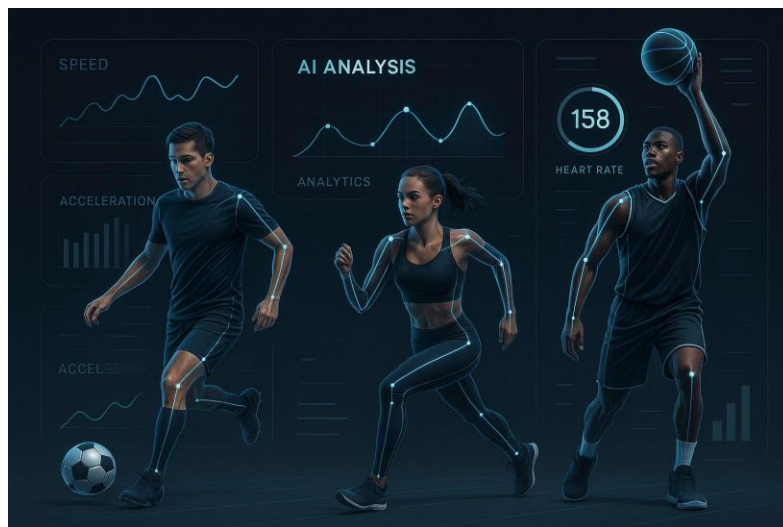
The reason is that live sports analysis requires not only detecting movement but also understanding the movement of athletes and sports objects. Optical Flow can provide horizontal and vertical motion information and generate trajectories that are useful for further analysis.

The selected solution provides a practical balance between accuracy and computational cost because Lucas-Kanade tracks selected feature points rather than calculating dense motion across every pixel.

The approach can be further optimized by:

- Tracking only important feature points.
- Reducing unnecessary frame processing.
- Using image pyramids efficiently.
- Limiting the tracking region.
- Using GPU acceleration when available.
- Combining Optical Flow with object detection.

Therefore, the final engineering decision is to use **optimized Lucas-Kanade Optical Flow for detailed sports motion tracking**, while Motion Estimation can be used as a faster baseline or supplementary motion-detection method.



7. Broader Considerations

Sustainability

The use of efficient computer vision algorithms can reduce unnecessary computational requirements. Feature-based Optical Flow processes selected features instead of calculating dense motion for every pixel, which can contribute to more efficient processing.

Environment

Efficient processing can reduce computational resource consumption, particularly when the system is deployed continuously during long sports events. Optimization of the algorithm can therefore help reduce unnecessary hardware and energy usage.

Society

Automated sports analytics can provide objective information about athlete movement and performance. It can support coaches, analysts, athletes, and sports organizations in making data-based decisions.

Safety

Motion tracking can potentially contribute to athlete safety by identifying unusual movement patterns, sudden changes in direction, or abnormal motion. Such information could support future systems for injury-risk monitoring.

Ethics

The use of computer vision in sports must consider athlete privacy, consent, responsible data storage, and appropriate use of collected video information. Automated analysis should not be used unfairly or without considering its limitations.

Economics

Automated motion analysis can reduce the time required for manual video analysis. Once developed, the system can process large amounts of sports video and provide useful information to analysts.

Accessibility

Automated visual analysis can convert complex sports video into measurable information such as trajectories, displacement, and movement direction. This can make performance information easier to interpret.

Professional Responsibility

A computer vision engineer must ensure that the system is properly tested and that its limitations are clearly communicated. Performance claims should be based on measured experimental results rather than assumptions. The system should also be maintained and improved when new requirements or deployment conditions arise.

8. Conclusion

The proposed project investigated motion tracking in sports analytics using computer vision. Two approaches, Optical Flow and Motion Estimation, were considered and evaluated based on tracking accuracy, processing speed, computational cost, and suitability for live sports applications.

Motion Estimation provides fast processing and lower computational requirements. It is suitable for identifying moving regions when rapid processing is the primary requirement. However, it provides relatively limited information about the detailed direction and trajectory of individual movements.

Optical Flow provides richer motion information by estimating the movement of pixels or feature points between consecutive frames. Lucas-Kanade Optical Flow provides an efficient feature-based implementation that is suitable for tracking motion while reducing the computational requirements of dense Optical Flow.

Based on the requirements of accurate athlete and sports-object tracking together with practical real-time performance, optimized Lucas-Kanade Optical Flow is selected as the proposed solution.

The major limitation is that tracking performance can decrease because of occlusion, rapid camera movement, illumination changes, motion blur, and insufficient feature points.

Future improvements can include object detection, multi-object tracking, camera-motion compensation, Kalman filtering, camera calibration, and GPU acceleration. These improvements can make the system more robust and suitable for large-scale live sports analytics.

9. Student Reflection

This assignment helped me understand how fundamental computer vision concepts can be applied to a real-world sports analytics problem. I learned that selecting an algorithm is not based only on accuracy. Processing speed, computational cost, robustness, and application requirements must also be considered.

One important learning outcome was understanding the difference between motion detection and motion tracking. Motion Estimation can quickly identify regions that have changed between frames, while Optical Flow can provide more detailed information about the direction and magnitude of movement.

I also learned how different stages of a computer vision system are connected. Video acquisition, preprocessing, feature detection, motion estimation, tracking, trajectory generation, and performance evaluation must work together to produce a useful final system.

Implementing the approaches using Python and OpenCV provided practical knowledge about processing video frames and handling feature points. It also demonstrated that real-world computer vision systems can behave differently depending on video quality, camera movement, lighting, and object visibility.

If additional time and resources were available, I would improve the system by integrating an object detection model to identify individual athletes and sports objects. I would also implement multi-object tracking so that individual players could maintain their identities throughout the video.

Another improvement would be camera calibration so that pixel displacement could be converted into real-world distance and speed. GPU acceleration could also be introduced to improve real-time performance.

Overall, the assignment improved my understanding of the practical trade-off between computer vision accuracy and computational performance and helped me understand how these factors influence engineering decisions.

10. References

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7. Python Software Foundation. *Python Documentation*. Python programming language documentation.
8. NumPy Developers. *NumPy Documentation*. Numerical computing library documentation.
9. Matplotlib Development Team. *Matplotlib Documentation*. Visualization library documentation.
10. Any sports video dataset or video used for experimentation should be cited according to its original source and license.
11. Any external images used in the report should be acknowledged with their original source.
12. AI-assisted tools used during preparation of the assignment should be acknowledged according to the institution's academic-integrity guidelines.